

EMBEDDED SYSTEM

- An embedded system is an electronic or electromechanical system designed to perform a specific function
- It is a combination of both hardware and firmware (software)
- Egs: ATM, Smartcard readers, Anti-lock braking systems, Automatic navigation systems etc.

EMBEDDED SYSTEMS Vs GENERAL COMPUTING SYSTEMS

General computing

- ❖ Generic hardware + general purpose operating system for executing a variety of applications
- ❖ Contains a general purpose OS
- ❖ Applications are alterable by the end user
- ❖ Performance is the key deciding factor

Embedded systems

- ❖ Special purpose hardware + embedded operating system for executing a specific set of applications
- ❖ May or may not contain an OS
- ❖ Firmware of the embedded system is non alterable
- ❖ Application specific requirements are the key deciding factors

EMBEDDED SYSTEMS Vs GENERAL COMPUTING SYSTEMS

General computing

- ❖ Less or not at all tailored towards reduced operating power requirements
- ❖ Response requirements are not time critical
- ❖ Need not be deterministic in execution behaviour

Embedded systems

- ❖ Highly tailored to take the advantage of power saving modes
- ❖ Response requirements are highly time critical
- ❖ execution behaviour is deterministic for hard real time embedded systems

CLASSIFICATION OF EMBEDDED SYSTEM

Classification can be done based on:

- **Generation**
- **Complexity and performance**
- **Deterministic behavior**
- **Triggering**
- **Functional requirements**

CLASSIFICATION OF EMBEDDED SYSTEM

- **BASED ON GENERATION**

- **First Generation**
- **Second Generation**
- **Third Generation**
- **Fourth Generation**
- **Fifth Generation**
- **Sixth Generation**

CLASSIFICATION OF EMBEDDED SYSTEM

- **BASED ON COMPLEXITY AND PERFORMANCE**

- **Small Scale Embedded systems**
- **Medium Scale Embedded systems**
- **Large Scale (Complex) Embedded systems**

CLASSIFICATION OF EMBEDDED SYSTEM

- **BASED ON DETERMINISTIC BEHAVIOUR**

- Hard Real Time Embedded systems
- SoftReal Time Embedded systems

- **BASED ON TRIGGERING**

- Even Trriggered
- Time Triggered

CLASSIFICATION OF EMBEDDED SYSTEM

- **BASED ON FUNCTIONAL REQUIREMENTS**

- Stand alone embedded systems
- Networked Embedded systems
- Real Time Embedded systems
- Mobile Embedded systems

MAJOR APPLICATION AREAS OF EMBEDDED SYSTEMS

- **Automotive industry**
- **Banking and Retail**
- **Card Readers**
- **Computer Networking Systems**
- **Computer Peripherals**
- **Consumer Electronics**
- **Health Care**
- **Home Automation**
- **Household Appliances**
- **Measurement and Instrumentation**
- **Telecom**

PURPOSE OF AN EMBEDDED SYSTEM

- **Data Collection/Storage/Representation**
- **Data Communication**
- **Data (Signal) Processing**
- **Monitoring**
- **Control**
- **Application Specific User Interface**

ELEMENTS OF AN EMBEDDED SYSTEM

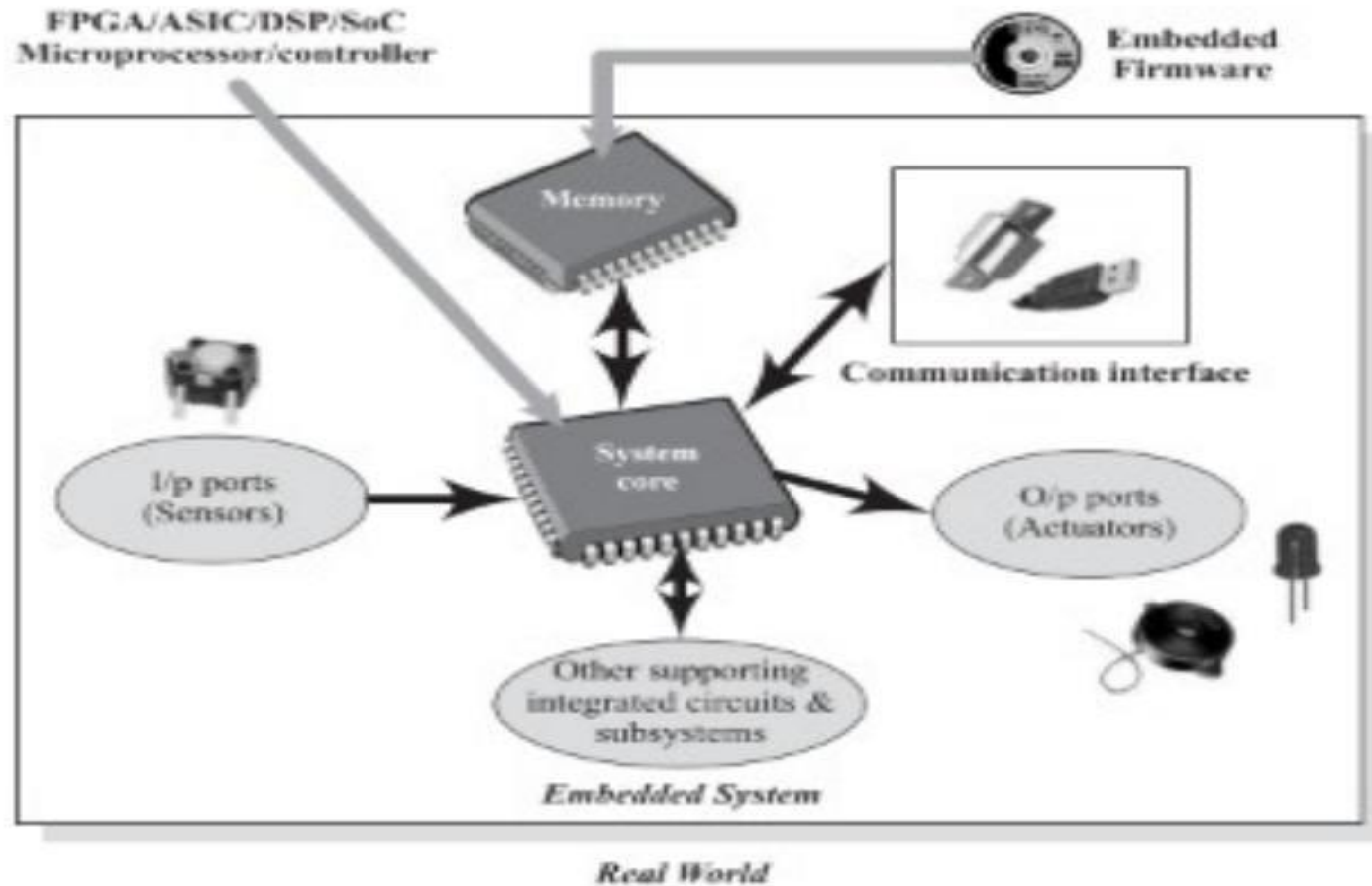


Fig. 2.1 Elements of an embedded system

Core of the EMBEDDED SYSTEM

1. General purpose and domain specific

a) Microprocessors

b) Microcontrollers

c) Digital Signal Processors

2. Application Specific Integrated circuits **ASICs**

3. Programmable Logic Devices **PLDs**

4. Commercial Off-the Shelf Components **COTS**

Core of the EMBEDDED SYSTEM

❖ Microprocessors

- CPU = ALU + CU + Registers
- Dependent unit
- System Architectures : Harvard & Von-Neumann
- Instruction set architectures : RISC & CISC
- Egs: Intel 4004, 8080, 8085, 8086, Pentium , Motorola 6800, Zilog Z80 etc

Core of the EMBEDDED SYSTEM

❖ Microcontrollers

- CPU + RAM + Special/General purpose registers + on chip ROM/ FLASH memory + timer & interrupt control units + I/O ports
- Independent unit
- System Architectures : Harvard & Von-Neumann
- Instruction set architectures : RISC & CISC
- Egs: Intel 8051 (general purpose controller), Atmel's AVR (Application Specific Controller for automotive domain.)

Core of the EMBEDDED SYSTEM

❖ Digital Signal Processors

- 8/16/32 bit microprocessors to meet computational demands and power constraints
- 2 to 3 times faster than general purpose microprocessors
- Perform real time calculations (SOP, Convolution, FFT, DFT etc)
- Implement algorithms in hardware which speeds up execution (general purpose processors implement algorithm in firmware , execution speed depends on the clock for the processor)
- DSP= Program memory + Data memory + Computational engine +I/O unit
- Applications : Audio Video Signal processing, Telecommunication, multimedia etc.
- Egs: Blackfin processors

Core of the EMBEDDED SYSTEM

❖ Application Specific Integrated Circuits

- Microchip designed to perform specific application
- Integrates several functions into a single chip
- Consumes very small area
- Pre-fabricated for a special application or can be custom fabricated for particular customer application
- Profitable only for large volume commercial production
- Require non refundable initial investment NRE charge (Non-Recurring Engineering)
- Eg: ADE7760 by Analog devices is an example for ASSP (Energy meter ASIC)

Core of the EMBEDDED SYSTEM

❖ Programmable Logic Devices PLDs

- Logic devices can be a) Fixed or b) programmable
- Fixed logic functions are permanent , cannot be altered
- Programmable logic devices can be re-configured
- No NRE costs
- Major types of PLDs are FPGA & CPLDs
- FPGAs offer high amount of logic density eg :Xilinx Virtex (Applications: Data processing and storage, instrumentation, telecommunication etc.)
- CPLDs offer smaller amounts of logic density eg: Xilinx CoolRunner (Applications : Critical control applications, portable applications)

Core of the EMBEDDED SYSTEM

❖ Commercial Off-The Shelf components

- COTS is used in its original state without any customizations
- Ready made products
- Easy integration and interoperability with existing system components
- No NRE costs
- Egs for COTS hardware units : remote controlled toy car control unit, high BW ADCs etc
- Eg for COTS product :TCP/IP plug-in module
- Identify the COTS for your system and give the plug-in option according to the specification
- Restrict the end user to stick to a vendor
- If the COTS manufacturer withdraws the product, commercial manufacturer of the embedded system using that COTS will be affected

MICROPROCESSORS Vs MICROCONTROLLERS

MICROPROCESSORS

- CPU = ALU + CU + Registers
- Dependent unit
- Mostly General purpose in design
- Does not contain Built-in I/O ports
- Targeted for systems where performance is important
- Limited power saving features

MICROCONTROLLERS

- CPU + RAM + Special/General purpose registers + on chip ROM/ FLASH memory + timer & interrupt control units + I/O ports
- Independent unit
- Application oriented or domain specific
- Contain Built-in I/O ports
- Targeted for Embedded systems where performance is not critical
- Includes lot of power saving features

RISC Vs CISC Processors / Controllers

RISC – REDUCED INSTRUCTION SET COMPUTING

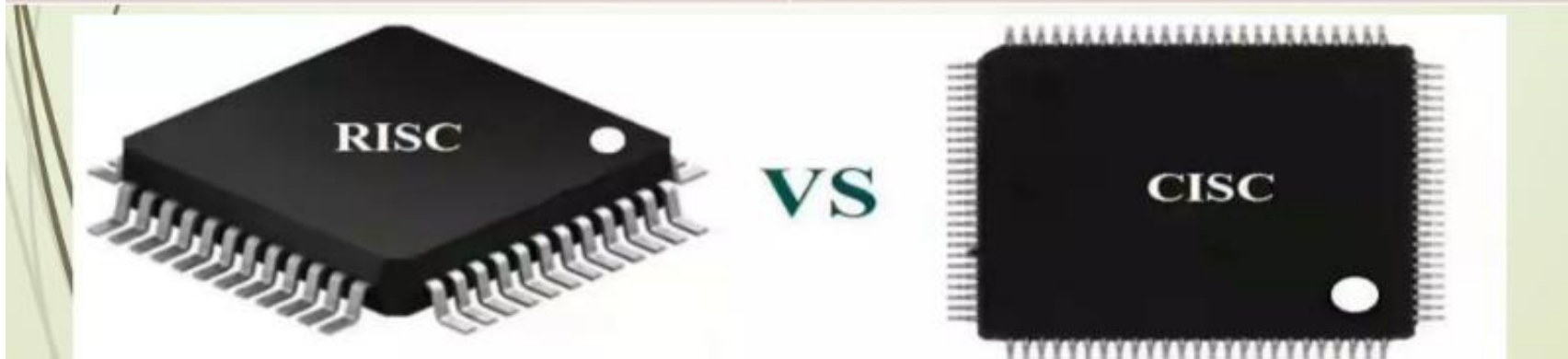
CISC – COMPLEX INSTRUCTION SET COMPUTING

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RISC	CISC
Lesser number of instructions	Greater number of Instructions
Instruction pipelining and increased execution speed	Generally no instruction pipelining feature
Orthogonal instruction set	Non-orthogonal instruction set
Operations are performed on registers only , the only memory operations are load and store.	Operations are performed on registers or memory depending on the instruction.
A large number of registers are available.	Limited number of general purpose registers.
Programmer needs to write more code to execute a task since the instructions are simpler ones.	Instructions are like macros in C language . A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC.

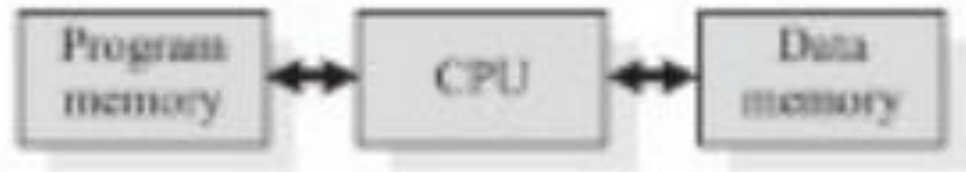
RISC Vs CISC Processors / Controllers

RISC	CISC
Single, fixed length instructions	Variable length instructions
Less silicon usage and pin count	More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.
With Harvard Architecture	Can be Harvard or Von-Neumann Architecture

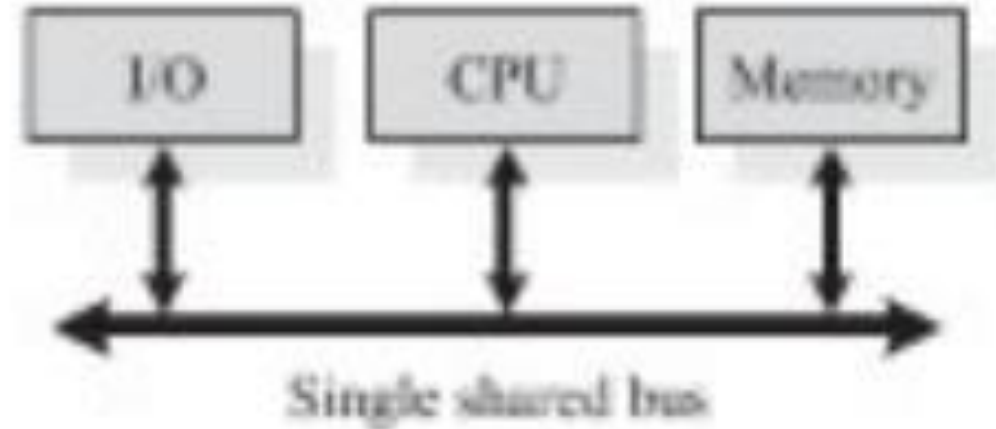


HARVARD Vs VON-NEUMANN ARCHITECTURES

HARVARD



VON-NEUMANN



HARVARD Vs VON-NEUMANN ARCHITECTURES

HARVARD

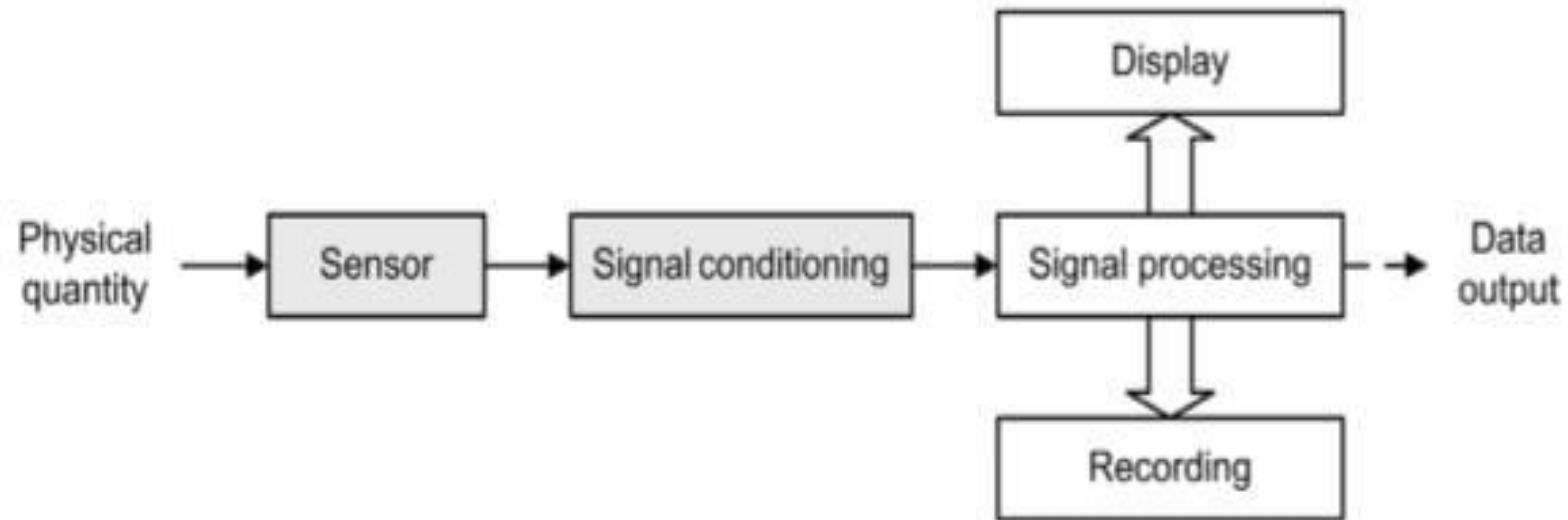
- Separate buses for instruction and Data fetching
- Easier to pipeline, so high performance can be achieved
- Comparitively High cost
- No memory alignment problems
- Data memory and Program memory are stored physically in different locations, so no chances for accidental corruption of program memory

VON-NEUMANN

- Single shared bus for instruction and Data fetching
- Low performance
- Cheaper
- Allows self modifying codes
- Data memory and Program memory are stored physically in same chip, chances for accidental corruption of program memory

SENSORS AND INTERFACING

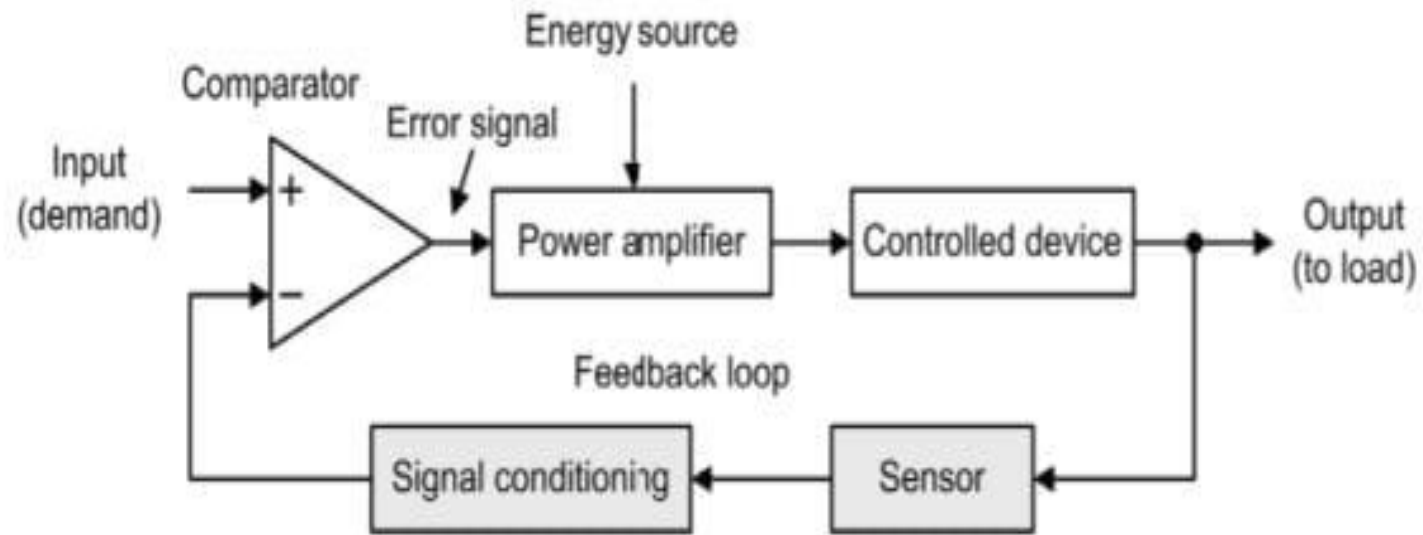
- INSTRUMENTATION SYSTEM



(a) An instrumentation system

SENSORS AND INTERFACING

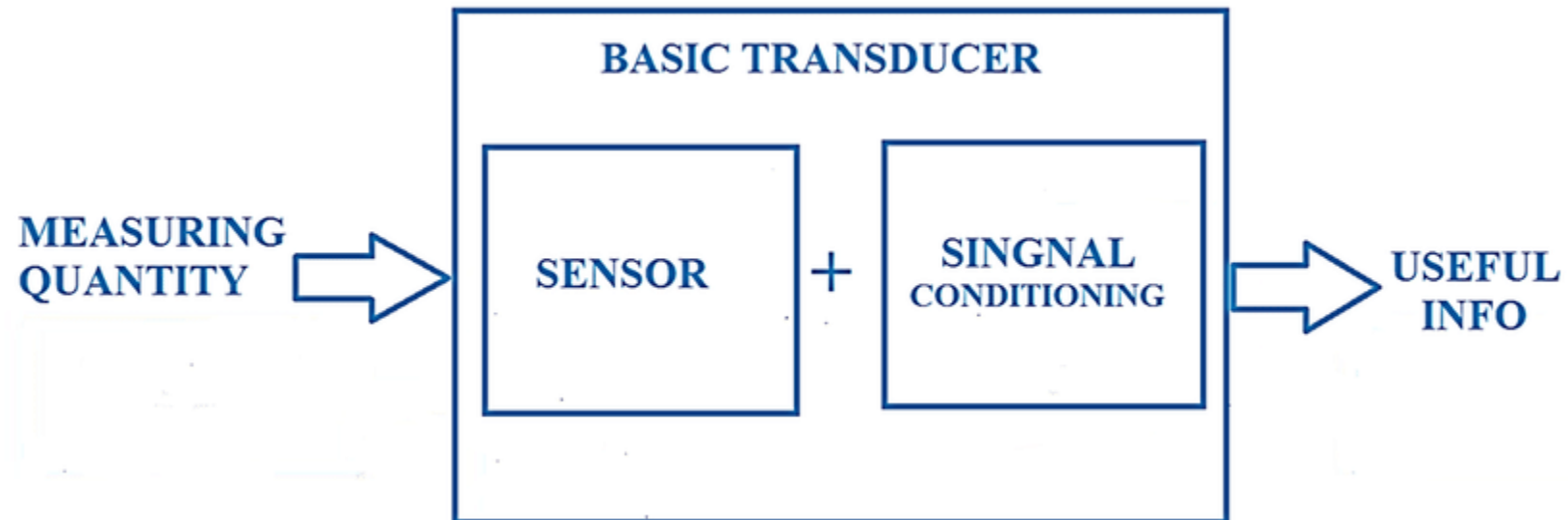
- **CONTROL SYSTEM**



(b) A control system

SENSORS AND INTERFACING

- **A BASIC TRANSDUCER** – works on the principle of TRANSDUCTION – converting one form of energy into another form (Electrical energy into any of the physical quantities or vice versa)



SENSORS AND INTERFACING

• APPLICATIONS OF TRANSDUCERS

- **Measurement field** – measuring physical parameters like temperature, light, pressure etc – thermocouples, microphones, photodiodes etc
- **Medical field** – ultrasound machines, blood pressure monitors, ECG machines, glucose meters etc
- **Automotive Systems** – monitoring engine performance, tire pressure, fuel levels etc
- **Aerospace** – Altitude measurement, engine monitoring in aircrafts, space crafts etc
- **Robotics & Automation Systems** – position sensing, force sensing, object detection, feedback n control etc

SENSORS AND INTERFACING

- **SENSORS**

- A sensor is a transducer device that converts energy from one form to another for any measurement or control purpose.
- Eg: the sensor which measures the distance between the cushion and magnet in the smart running shoe is a magnetic hall effect sensor.

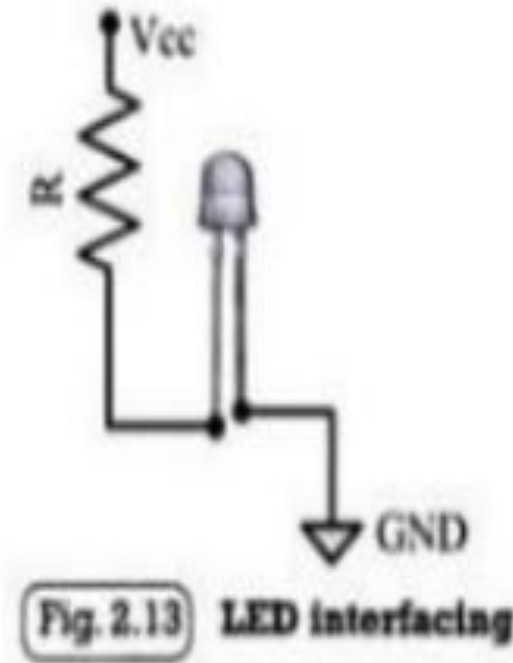
- **ACTUATORS**

- Actuator is a form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device.
- Eg: In the smart running shoes, the actuator used for adjusting the position of the cushioning element is a micro stepper motor

SENSORS AND INTERFACING

- **LIGHT EMITTING DIODE (LED)**

- Light Emitting Diode is a p-n junction diode and it contains an anode and a cathode.
- For proper functioning of the LED, the anode should be connected to positive terminal of the supply voltage and cathode to the negative terminal of supply voltage.
- Light Emitting Diode (LED) is an important output device for visual indication in any embedded system.



SENSORS AND INTERFACING

- **7- SEGMENT LED DISPLAY**

- The 7-segment LED display is an output device for displaying alpha numeric characters.
- It contains 8 light-emitting diode (LED) segments arranged in a special form. Out of the 8 LED segments, 7 are used for displaying alpha numeric characters and 1 is used for representing 'decimal point' in decimal number display.

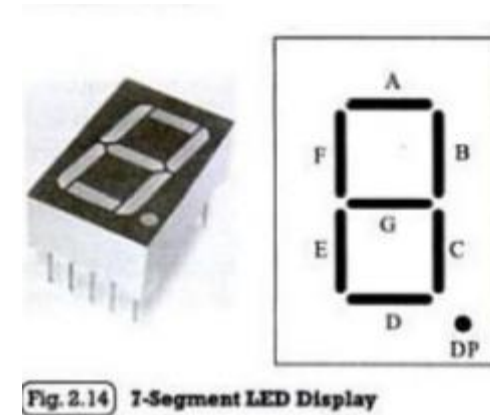


Fig. 2.14 7-Segment LED Display

SENSORS AND INTERFACING

- Common anode and common cathode configurations of a 7-segment led display

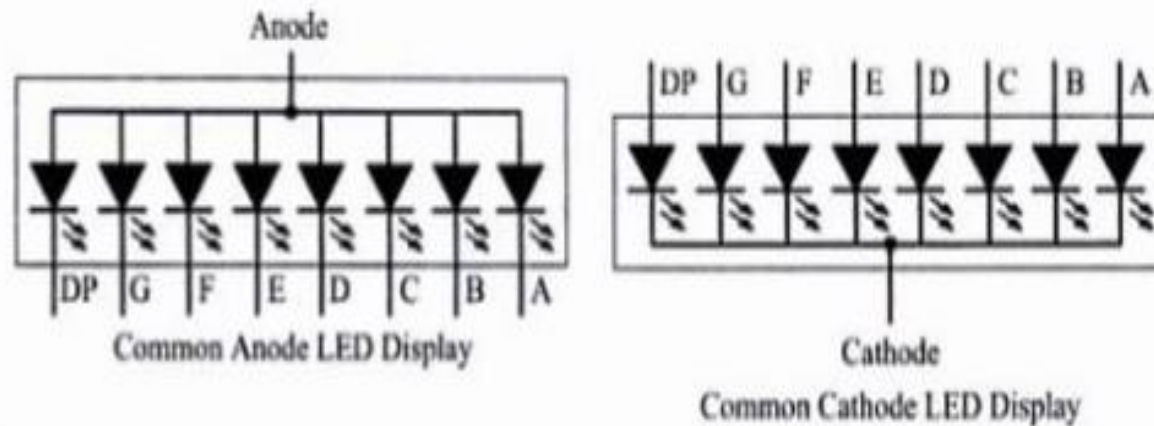


Fig. 2.15 Common anode and cathode configurations of a 7-segment LED Display

THANK YOU.....

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